Comprehending Survey Questions: Basic Processes

SurvMeth 632

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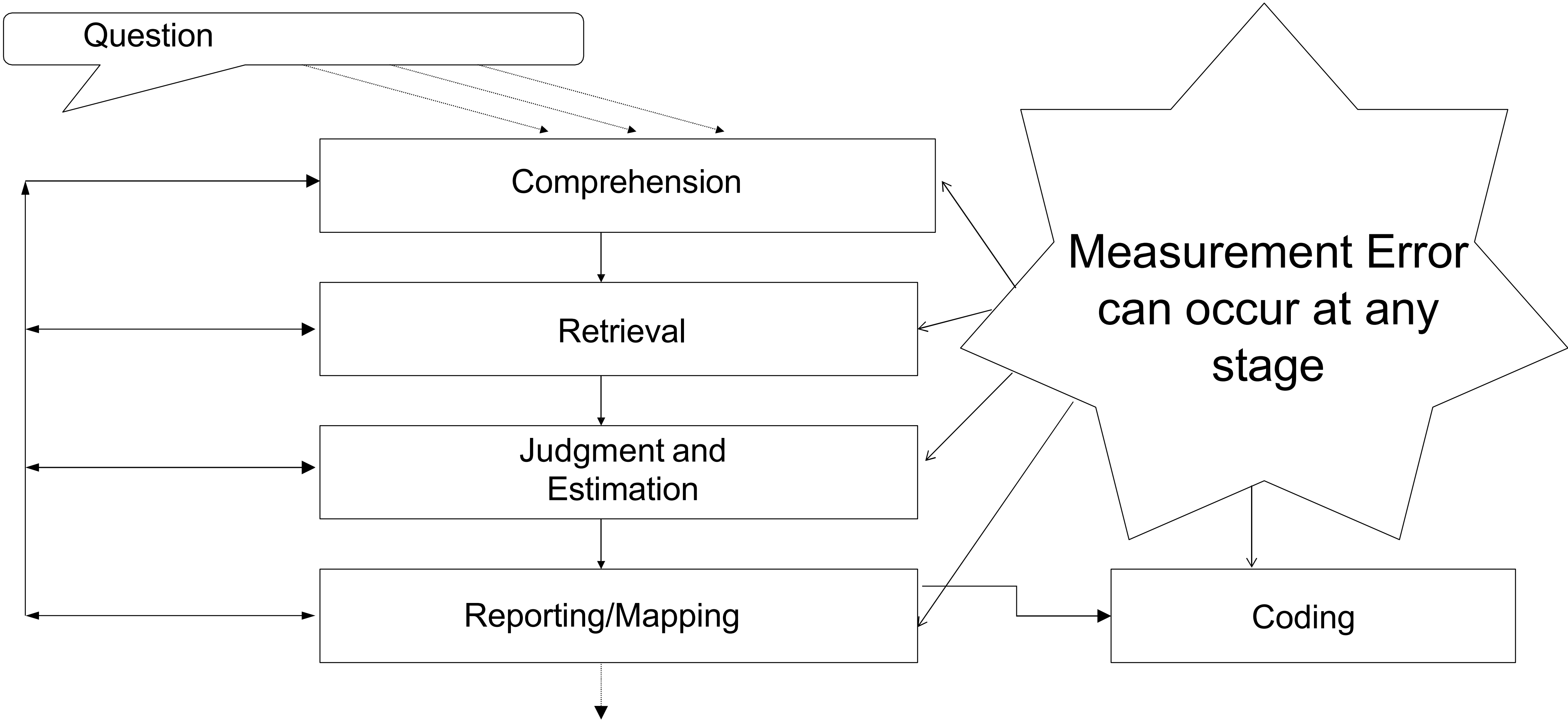

Question
Comprehension
Retrieval
Judgment and Estimation
Reporting/Mapping
Measurement Error can occur at any stage
Coding

Other Models of the Response Process

- Two-track models
  - Cannell's Process Theory
    - Systematic versus superficial processing
  - Satisficing Model
    - Satisficers versus optimizers
  - Strack and Martin's Two-Track Theory
    - Existing judgment versus derived judgment

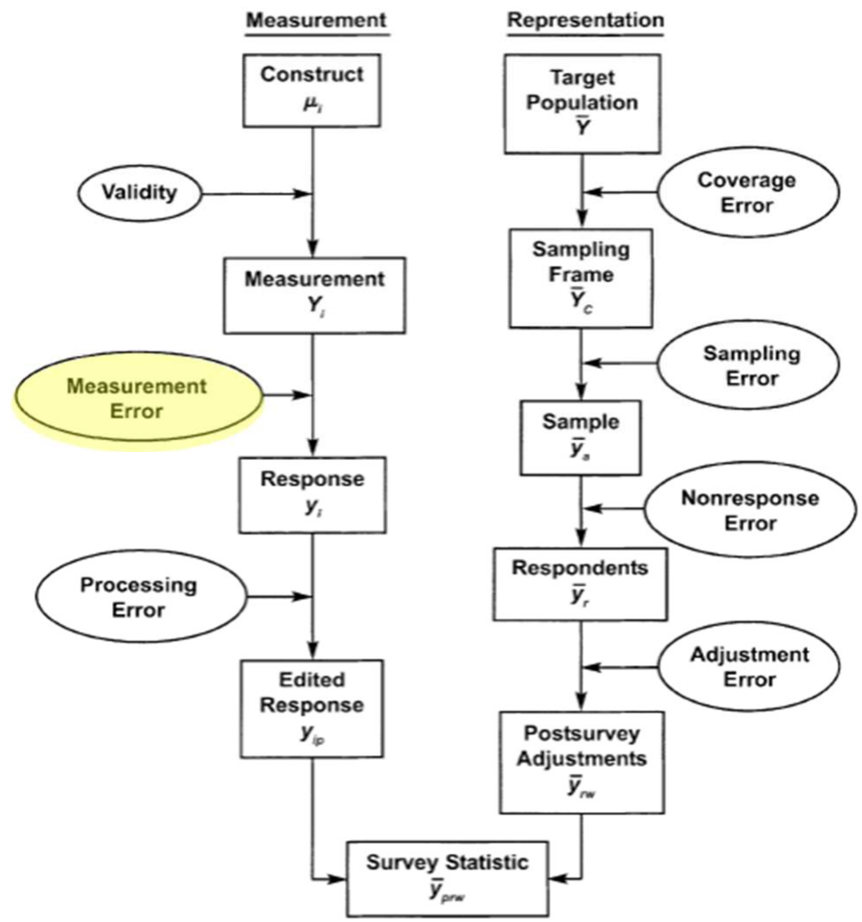
- These models make useful distinctions that help determine the response process that is used
- Four step model is more descriptive about the entire response process
- The two tracks in these models may be more blurry than distinct paths.

Cognitive Aspects of Survey Methodology (CASM) movement

- Much of the research for this course and applications of cognitive psychology is the result of a collaboration between cognitive psychologists and survey methodologists that began in the 1980s.
- Cognitive psychology provided better understanding of the response process for survey methodologists.
- Survey methodology provided real life applications to develop and test cognitive theory on phenomena like memory of everyday events and how people fill in or estimate incomplete information

Survey Error

This course



Bias Versus Variance

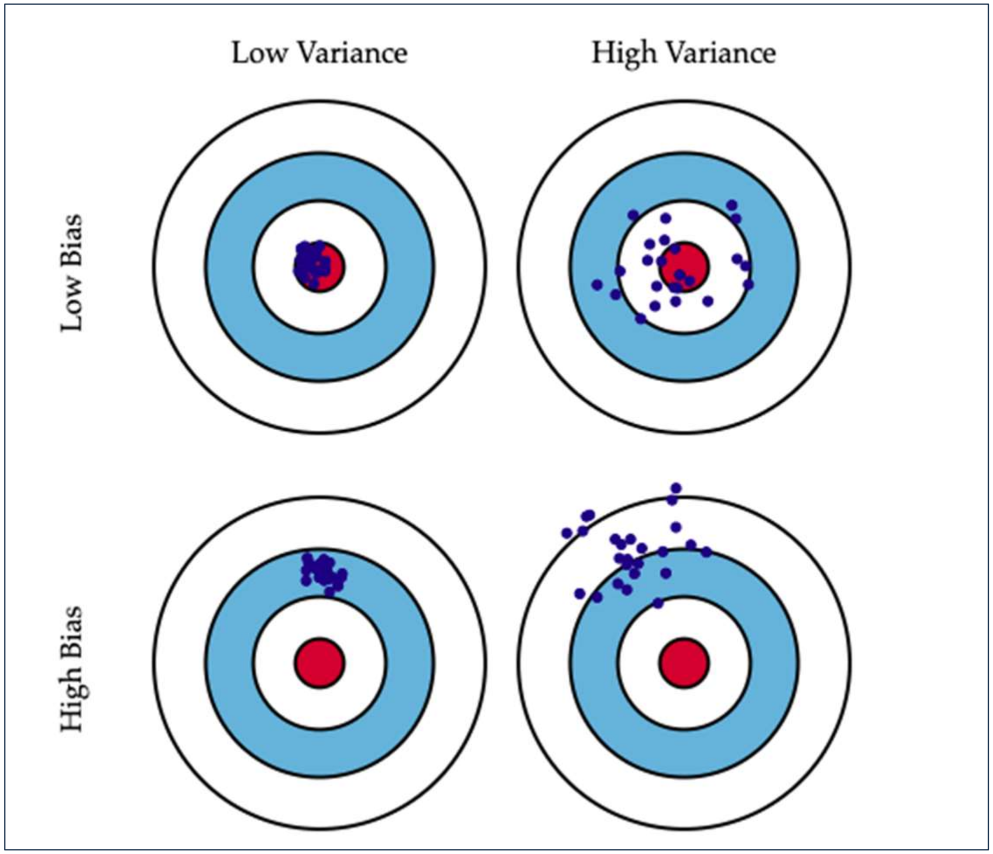


Image source: https://scott.fortmann-roe.com/docs/BiasVariance.html

Measurement Error

- Individual survey responses may depart from the *true value* –
  – This difference is measurement error
  – Although the objective truth may be unknowable
- Disciplinary differences in how think about measurement error:
  – Statistical: $y_{jt} = X_j + \rho_{jt}$
  – Psychometric:
    - Reliability – same results for > 1 test?
    - Validity – are the results accurate
  – Cognitive: discrepancy due to misunderstanding, forgetting, recency, etc.
- but all share concern that inaccurate responses undermine survey estimates
- and struggle with how to know the "truth"

Outline: Comprehending Survey Questions

1. Extracting meaning from Questions
   a) Literal vs. non-literal meaning
   b) Literal meaning (semantics)
   c) Non-literal meaning (pragmatics)

2. Grounding Question Meaning: the Standardization Debate
   a) Standardized Interviewing
   b) Conversational interviewing
   c) Empirical comparisons of the two techniques

Literal versus non-literal meaning

- *Representation of* the sentence (question)
  - Specification of the underlying grammatical and logical structure of a sentence
  - Meanings of words (in mental dictionary) relatively stable and conventional
  - *Semantics*: word meanings and their combination
- *Representation about* the sentence (question)
  - People infer meaning beyond the (conventional) meanings of words
    - *response categories* lead *R*s to infer intensity or severity, e.g. How often are you depressed? Once a week…. Every day" vs. "Once a year ... Once a week."
    - previous *Q*s and answers may affect interpretation of current *Q*
    - *R*'s beliefs about purpose of survey may affect interpretation of *Q*
  - Inferences vary between *R*s, less conventional than word meaning
  - *Pragmatics*: language use; <u>inferences</u> about intended meaning

Question Syntax

- Often simple to move between an interrogative and a declarative sentence particularly for yes-no questions.
  - Have you had a mortgage on this property since the first day of June?
  - You have had a mortgage on this property since the first day of June.
  - Easy to identify the focus of this question

- More complicated with wh-words
  - What would you have to spend each month in order to provide the basic necessities for your family?
  - I would have to spend t* each month in order to provide the basic necessities for my family.

- How about this example?
  - When did Lydia tell Emily the phone would be fixed?

Processing of questions beginning with Wh-words

- With *wh-words* people store in memory information predicting they will encounter a missing part of the sentence
- Who(m), which, and what often begin components of a sentence called arguments that are required by verbs and prepositions.
- Where, when, why, and how often begin components called adjuncts that are optional parts of the question so it is harder to find the missing part of the sentence.

Complexity

- Complex questions place burden on working memory.
  - During the past 12 months, since January 1, 2024, how many times have you seen or talked to a doctor or assistant about your health? Do not count any time you might have seen a doctor while you were a patient in a hospital, but count all other times you actually saw or talked to a medical doctor of any kind.
  - Some people feel the federal government should take action to reduce the inflation rate even if it means that unemployment would go up a lot. Others feel the government should take action to reduce the rate of unemployment even if it means the inflation rate would go up a lot. Where you place your self on this [7 point] scale?

- Consequences of overburdening working memory
  - Items drop out of working memory
  - Cognitive processing slows down

Uncertainty Space

- Questions have an uncertainty space that represents a set of possibilities that represents an answer.

- Questions can be modified to clarify the uncertainty space.
  - What time did Calvin usually leave home to go to work last week?
  - What time did Calvin usually leave home to go to work last week?
    - 7 AM-11:59 AM
    - 12 PM- 4:59 PM
    - 5 PM-10 PM
  - What time in the morning did Calvin usually leave home to go to work last week?

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Literal Meaning: Conceptual Variability

- The same word means different things to different people

  Do you think children suffer any ill effects from watching programmes with violence in them, other than ordinary Westerns?

- Belson (1981) determined that *R*s interpreted *children*, *ill effects* and *violence* in numerous ways
  - "children": < 8 years, < 19 - 20 years
  - children as students
  - only 8% interpreted all words in question as intended

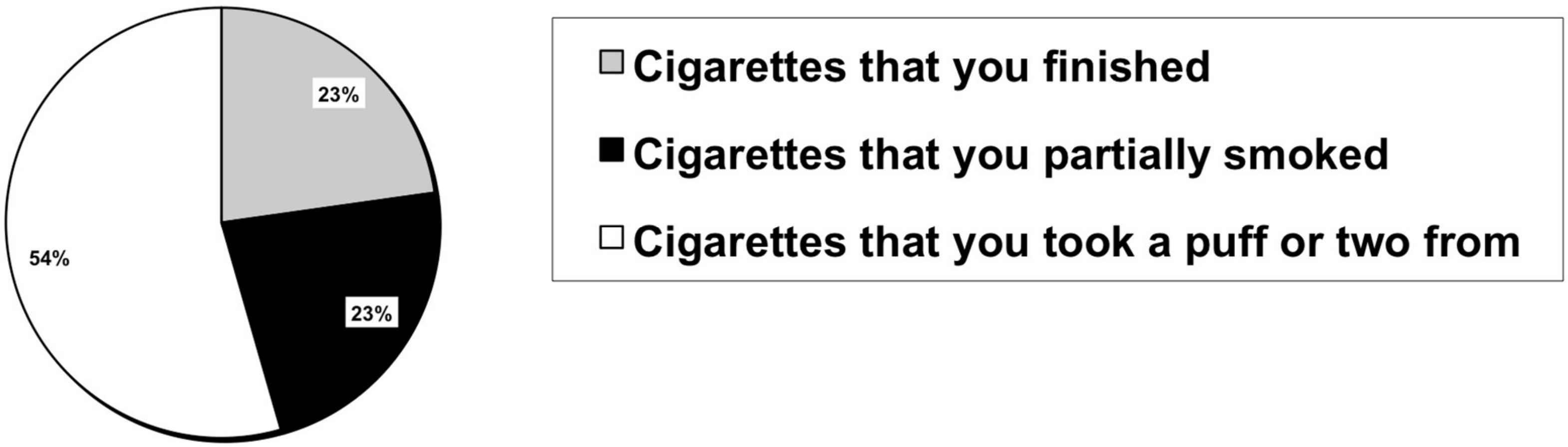
❖ Caveat: We don't know if this led to incorrect responses

- For someone who thinks *people* suffer no ill effects from watching programs with violence, misinterpreting *children* won't affect response

Conceptual Variability (2)

- Schober, Suessbrick & Conrad (2018)
  - CPS Tobacco Use Supplement interview followed by interpretation post-test

In your entire life, have you smoked at least 100 cigarettes?



Interpretation of cigarette

  - Most frequent interpretation held by only 54%, so interpretation differed between *R*s

Conceptual Variability (3)

- Schober et al., 2018 (cont'd)
  - Differences in interpretation affect response "accuracy"
  - After interview and interpretation post-test, *R*s answer original items again (on paper, i.e., no *Iwer*)
  - ½ *R*s also given a standard definition; other ½ given no definition
  - 5.3% of *R*s *change answers* (compared to original answer) in no definition condition
  - 12.8% *R*s *change answers* when given definition, suggesting initial interpretation – and response – was incorrect (with respect to definition)

Ambiguity

- Lexical

  The best way to prevent cancer is to catch it early

  strongly agree   somewhat agree   somewhat disagree   strongly disagree

  – Unclear, at least initially, which sense of "catch" is intended
    - "Become ill with" or "detect"

  – Word has more than one meaning
- Mapping

  I: Last week, did you have more than one job including part-time, evening or weekend work?

  R: Um...I babysit for two families. Is that one job or two?

  – Unclear how to apply (map) "more than one job" to one's circumstances

  – What should be *included,* and what should be *excluded*?

Beyond Literal Meaning: Presupposition

- Some sentences (questions) refer to what speaker feels listener knows or believes, either from previous conversation or just because speaker feels listener would know or believe this
- This is *presupposed*
- What if presupposition is not accepted by R?

  Family life often suffers because working parents concentrate too much on their work. Do you strongly agree, somewhat agree, etc.?

  – What is presupposed?
  – Does offering a "don't know" option solve the problem?

Presuppositions

- Presuppositions limit the uncertainty space

- Many presuppositions are necessary in a questionnaire, but some can make it hard for respondents to answer

- What does this question presuppose?
  - How often do you do MODERATE-INTENSITY LEISURE-TIME physical activities?
  - *If necessary, prompt with: How many times per day, per week, per month, or per year do you do these activities?*

- Attitude questions may presuppose that respondents have an opinion

- Filter questions and don’t know options can help

Beyond Literal Meaning: Five Principles of Language Use (Clark and Schober, 1992)

- Language has to do with people and what then mean not just what words mean.
- Interviewers and respondents engage in a social process
  - **Principle of speaker's meaning**: addressees recognize what the speakers mean by what they say and do (e.g., Toronto Law to Protect Squirrels Hit by Mayor)
  - **Principle of utterance design**: speakers design each utterance so addressees can figure out what they mean by considering their utterance against their personal and cultural <u>*common ground*</u> *(e.g., common ground with other graduate students)*
  - **Principle of accumulation**: participants add to their common ground each time they contribute to it successfully (bridging inferences between given and new information)
  - **Cooperative principle**: conversational contribution is made as required, at the stage it occurs, by the accepted purpose or direction of the exchange
  - **Principle of grounding**: participants try to reach the mutual belief that the addressees have understood what the speaker meant (speakers collaborate to reach intended meanings of utterances)

What’s different about survey interviews (Clark and Schober, 1992)

- Interviewers are intermediaries between the researcher and the respondent

- Interpretability presumption: respondents assume researcher chose words carefully
  - Respondents answer vaguely worded questions in idiosyncratic ways (e.g., children, few)
  - Respondents fail to see when the researcher is using words differently from the way they use them
  - Respondents seem to have stable opinions on things they know nothing about

- Questions have a perspective (e.g., How fast were the cars going when they smashed into each other?)
  - Loaded questions set the perspective from which questions are to be answered (e.g., government revenues versus taxes)
  - Response alternatives to a question help determine the domain of inquiry in which it is to be answered (e.g., forbid versus allow public speech against democracy)
  - Questions with and without response alternatives imply different perspectives (e.g., What is the most important problem facing the country today?)
  - Response alternatives construed as typical in the population (e.g., changing response options for question about TV viewing)

What’s different about survey interviews? (Clark and Schober, 1992)

- Pressure to respond: answer is given in the time an answer is expected to be given
  - Respondents estimate factual answers that would take too long to figure out precisely (e.g., telescoping)
  - Responses to survey questions change with mode of administration (Telephone is the fastest)

- Pressure for consistency
  - Knowledge filters can suppress opinions in later questions
  - People use earlier answers as evidence for later judgments

- Quest for structure
  - People interpret successive questions as related in topic unless they are told otherwise
  - General questions following specific questions can have inclusive or exclusive interpretation (e.g., dating and happiness with life)
  - Explicitly state connections take precedence (e.g., preface indicates two questions are independent)

Beyond Literal Meaning: Common Ground

(H.H. Clark, e.g., 1996)

- Listeners go beyond literal meaning based on their knowledge of the speaker and what
  - has been said up until now
  - has been presupposed
  - the “other” can be reasonably assumed to know
  - inferences about speaker’s intentions have been made
    - bridging inferences (Clark ‘75)

    I met a man yesterday. The bastard stole all my money.

    Alex went to a party last night. He is going to get drunk *again* tonight.
- Common ground ≈ working memory

Beyond Literal Meaning: Researcher’s affiliation can affect how *R* interprets *Q*s

- Norenzayan and Schwarz (1999)
  - asked *R*s to explain a case of workplace homicide (open text)
  - The questionnaire was printed on either letterhead of “Institute for Personality Research" or "Institute for Social Research”
  - *R*s’ explanations emphasized personality or social-contextual variables depending on whether the sponsor’s affiliation implied they did personality or social research
- Galesic and Tourangeau (2007)
  - Asked two groups of *R*s identical questions about the same workplace behavior in either
    - alleged “Sexual Harassment Survey” conducted for Women Against Sexual Harassment
    - alleged “Work Atmosphere Survey” conducted for Work Environment Institute
  - *R*s perceived same behaviors in first “survey” as more likely to represent sexual harassment
- These inferences (e.g., about sponsorship) usually unwarranted

Grice's Cooperative Principle

- Listeners infer what speaker means by assuming statements are intended to promote communication, i.e., speaker is being cooperative
  - A: *I am out of gas*
  - B: *There is a garage around the corner*
  - o B infers that A is asking for information about location of gas station (otherwise, would be uncooperative to say it)
- Cooperative Principle comprised of 4 "maxims" (rules) that listeners assume speaker is following
  - Quantity (say as much as necessary but not more)
  - Quality (say what you believe to be truthful)
  - Relation (make contributions relevant to discourse)
  - Manner (be clear)
- Intentionally violating (flouting) maxims:
  - A and B both know that X has divulged a secret about A
  - A: *X is a fine friend.*

Question Order Effects

- Info in common ground available for answering current *Q*
  - Includes info on which previous answer is based
  - Highly accessible (in working memory) when answering current *Q*
  - If topics of previous and current *Q* are related, natural to reuse info used for previous answer
- But *R*s may also recognize (infer) that reusing info on which previous answer is based is redundant and thus may not be *cooperative*
  - Redundancy violates maxims of quantity and relation
  - If *R* recognizes redundancy, may exclude the considerations on which previous answer is based from their formulation of current answer
  - rendering the two answers unrelated
- But if unaware of redundancy, *R* may automatically include info on which previous answer was based (and which in common ground) when answering current *Q*
  - Leading to correlated responses
- Inclusion or exclusion not necessarily related to accuracy of response

Schwarz, Strack and Mai ('91)

Correlation of Relationship Satisfaction (specific) and Life-Satisfaction (general)

Condition	***r***
Specific-general	0.67*
Specific-general, with joint lead-in	0.18
Specific-general, explicit inclusion	0.61*
Specific-general, explicit exclusion	0.20

*p < .05

Standardization Debate: Outline

1. What is Standardized Interviewing?
2. Proposal for Conversational Interviewing
3. Empirical comparisons of the two techniques

What is standardized interviewing?

- Prevailing philosophy and practice of collecting survey data:
  - *Iwer*s expected to read question exactly as worded
    - If answer not acceptable, *I* must use *neutral probes*
  - This approach should, in principle, promote *comparable* data across *Iwer*s
    - different responses should reflect actual differences between *R*s, not differences in question wording (stimulus)
  - Standardize the stimulus so interaction is scripted
  - Statistical goal is to remove the *Iwer* as a source of error variance
    - “interviewer variance”
- Procedure/rationale articulated by Fowler and Mangione (‘90)

Pros and cons of standardized wording

- What's good
  - Comparability: Can be sure all *R*s receive same question stimulus
  - Should, in principle, eliminate *interviewer effects*
  - Leads to fast interviews and, thus, reduced interviewer costs
- What's bad
  - Prevents *grounding* – ability to talk about meaning – because this involves unscripted interaction
  - Inability to ground meaning increases chances of misunderstanding
  - Misunderstanding may lead to inaccurate answers

Proposal for Conversational Interviewing

- Based on on their analyses of small set of interactions in General Social Survey (GSS) interviews, Suchman & Jordan (1990), argued that *I*s should be empowered to use *conversational resources* to promote accurate answers
  - i.e., not just consistent responses
  - "conversational resources" include "grounding" – back-and-forth to establish that speaker and listener understand each other ("mutual intelligibility")
- Proposed that because strict standardization prevents conversational methods for establishing mutual intelligibility, can undermine response <u>validity</u>
  - though not necessarily <u>reliability</u>, i.e., *R*s answers may be consistently wrong

Grounding*

(H.H. Clark, e.g., 1996)

- Speakers and listeners indicate they believe they understand each other
  - "uh huh," "I see," "got you," nodding, other gestures
- If there is not mutual understanding, addressee can seek clarification from speaker
  - "What do you mean by …?"
- Collaborative view:
  - Grasping meaning requires "back and forth" between speaker and listener; they have grounded the speaker's meaning when they agree they understand each other <u>well enough for current purposes</u>
  - Clark & Schaeffer (1987) analyzed corpus of directory assistance calls; found ~82% of exchanges seemed problem-free but remainder involved extra dialog before listener "accepted" speaker's contribution

*Grounding ≠ Common Ground

Grounding (2)

O. Directory Enquiries, for which town, please?

C. In Cambridge

O. What’s the name of the people?

C. It’s the Shanghai Restaurant, it’s not in my directory, but I know it exists

O. It’s Cambridge 12345

C. 12345

O. That’s right

C. Thank you very much

O.Thank you, good bye

Clark & Schaeffer, 1987

Conversational Interviewing Philosophy and Procedure*

- Primary goal is to reduce mapping ambiguities
- *Iwer* authorized to say whatever it takes to be sure *R* understands *Q* as intended
  - “ground” understanding
- *Iwer* must initially read question as worded but then can clarify *Q*
  - paraphrase the question or provide a relevant definition
  - either when *R* specifically asks for clarificatio, or *Iwer* deems it necessary
- Pretesting cannot anticipate many mapping ambiguities in a diverse sample; clarification needed <u>in interview,</u> i.e., in real time when *R*s’ circumstances become evident to *Iwer*

*Schober & Conrad, 1997

Pros and cons

- What's good
  - Should promote the intended understanding of questions and, thus, high response accuracy
- What's bad
  - Clarification may take extra time
  - Critics worry that
    - *I*s may (mis)lead *R*s
    - Technique may increase interviewer variance

Conversational interviewing

(Conrad & Schober, 2000; Schober & Conrad, 1997; Schober, Conrad & Fricker, 2004)

- *I* and *R* work together to assure *R* understands question as intended
  - interviewer reads question as worded, then
  - interviewer says whatever is necessary to assure *R* interprets question as intended, i.e., to ground meaning
- Goal is to standardize meaning, *even if wording varies*

Empirical comparisons of the two techniques: Lab Experiment

- Schober and Conrad (1997)
  - two groups of interviewers both trained on survey concepts
    - one group trained in strictly standardized techniques
    - other group trained to use conversational (“flexible”) technique
  - Interviewed respondents (members of public) via telephone in laboratory
  - Asked 12 questions borrowed from on-going US Federal surveys
  - *R*s answer on basis fictional scenarios so that response accuracy could be determined

Example question

- *Has Kelly purchased or had expenses for household furniture?*

Straightforward scenario, example

KATZ'S

Furniture Mart

Brooks End Table 149.99

713000000075

Tax…….. 11.99

TOTL **161.98**

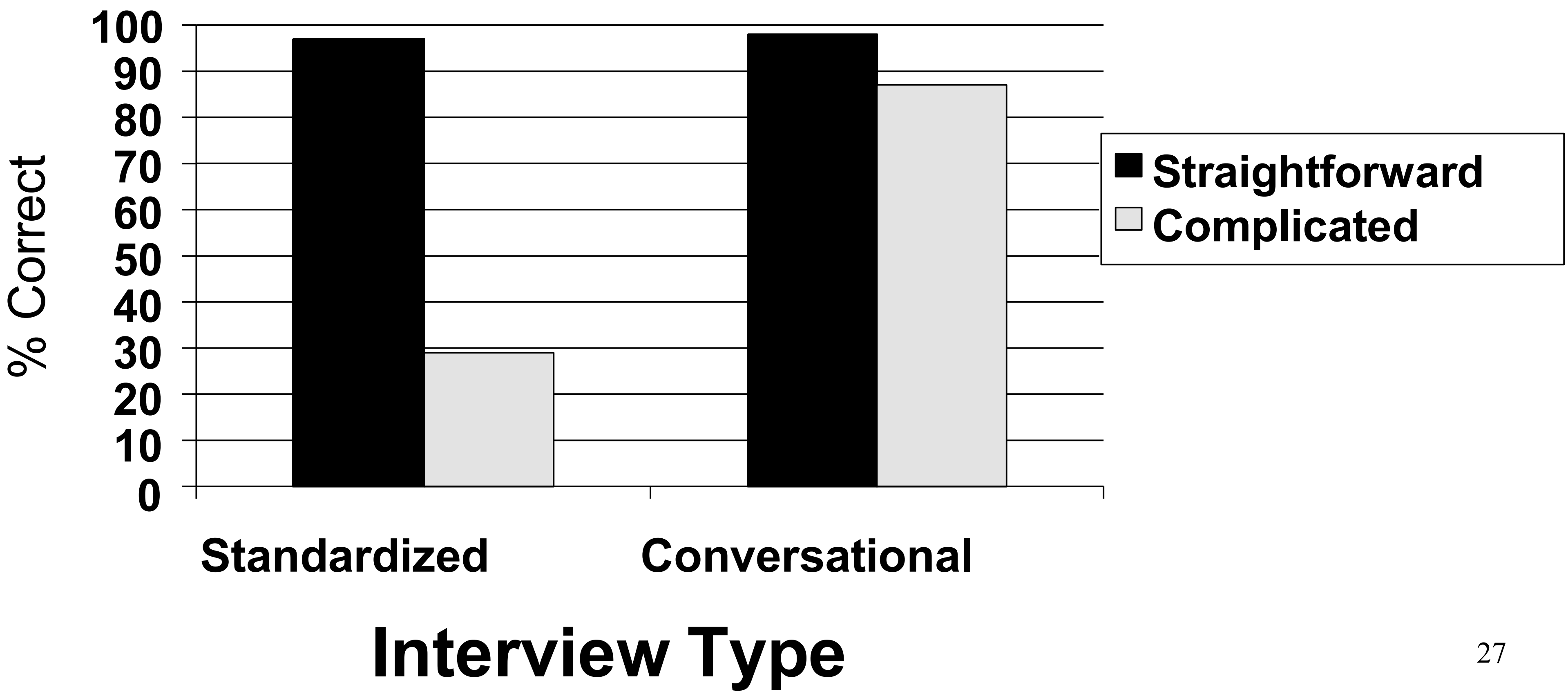
B112 882000002

4330 7:49 PM

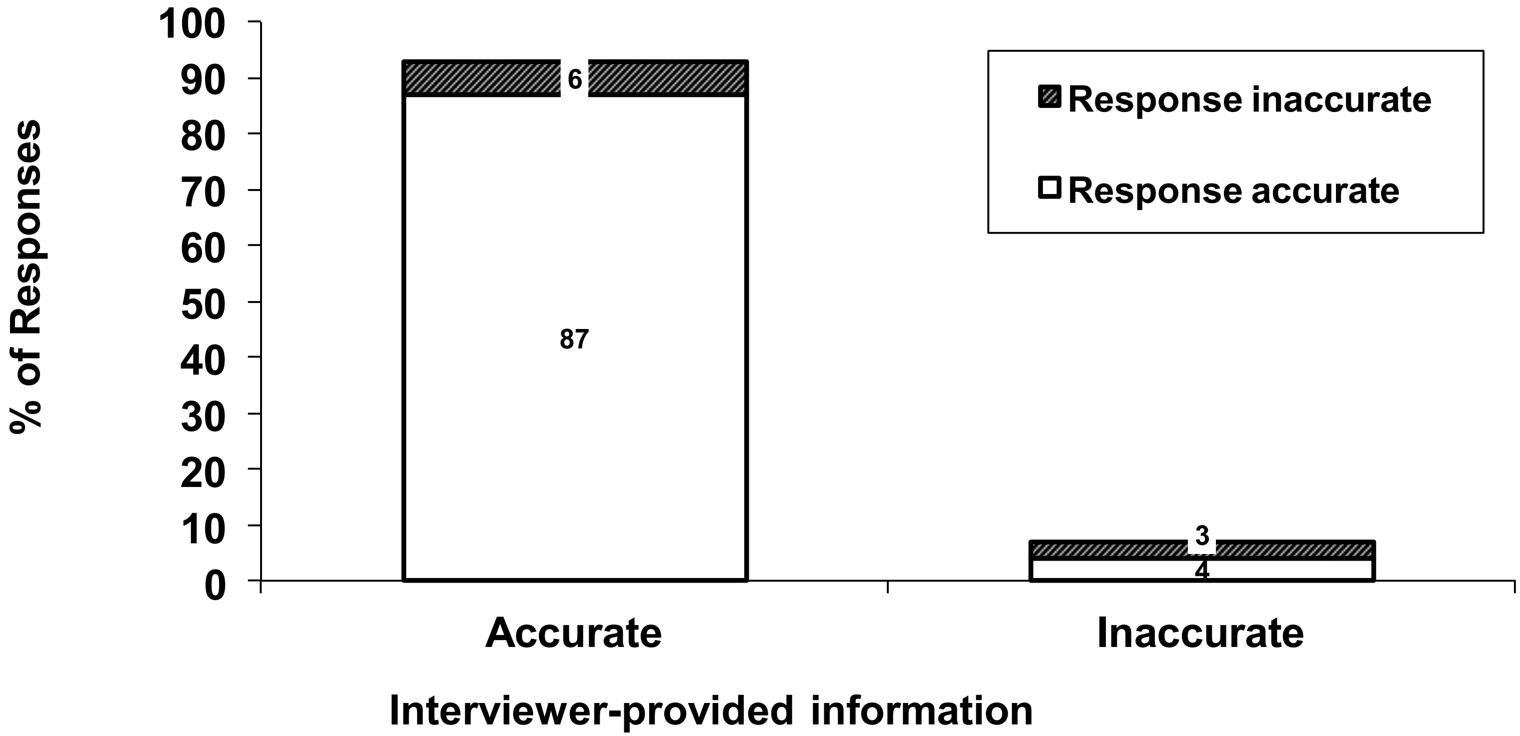
Complicated scenario, example

KATZ'S	
Furniture Mart	
Lumin Floor Lamp	149.99
	713000000075
Tax……..	11.99
TOTL	**161.98**
B112 882000002	
4330 7:49 PM	

Response Accuracy



Accuracy of Interviewer-provided Info



Minutes per Interview

	Standardized	Conversat'al
Median	3.41	11.47
Min	2.48	6.10
Max	5.99	35.44

- May overstate duration due to CI in practice because in experiment, ½ Qs involve complicated mappings; likely fewer in actual interviews

Field experiment

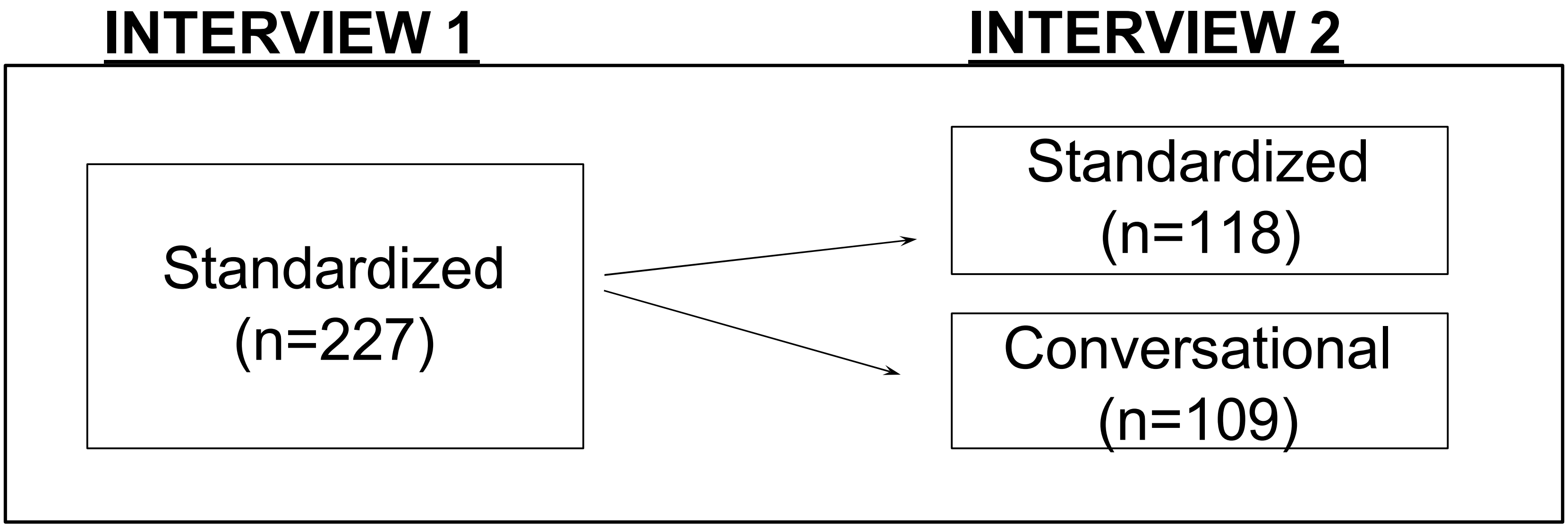
(Conrad & Schober, 2000)

- Hard to measure understanding in real survey settings (as opposed to laboratory settings)
  - Without access to respondents' circumstances, can't tell if responses fit official definitions
  - Can't directly measure response accuracy
    - record checks and diaries are difficult to carry out and expensive
    - Records, if they can be identified, may not be 100% accurate

Measuring comprehension

- We *can* determine if conversational interviewing *changes* *R*s' understanding
- Logic:
  - if conversational interviewing improves comprehension
  - then *R*s in <u>standardized</u> interviews should change their responses in a subsequent <u>conversational</u> interview *more* than they would in a subsequent <u>standardized</u> interview

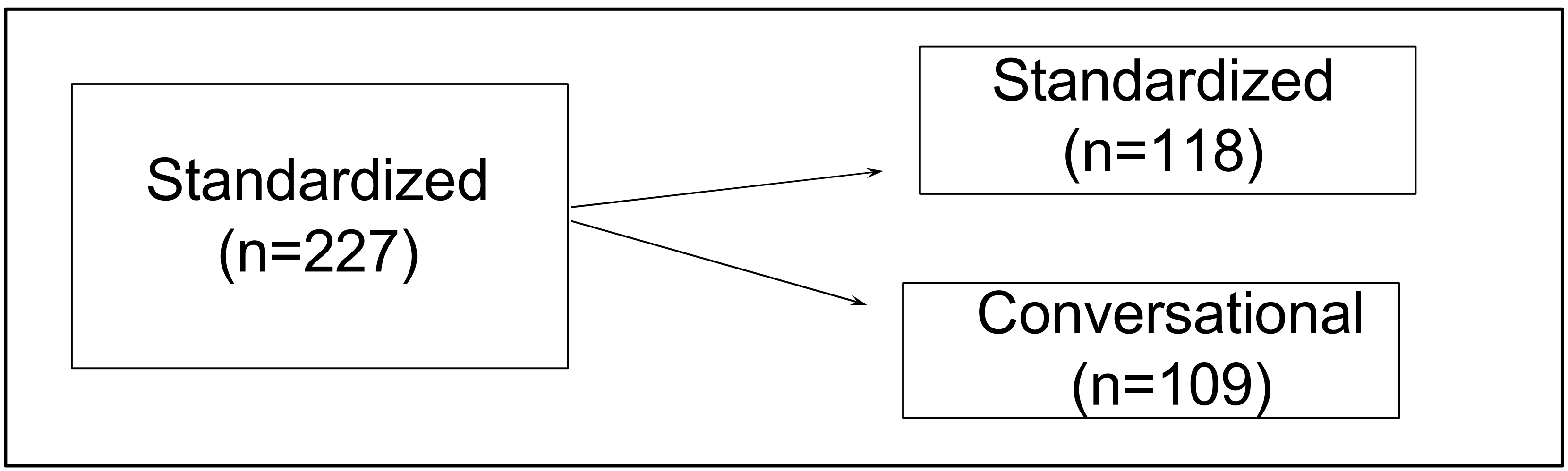
- Each *R* (n = 227) interviewed twice by different *Iwer*s asking same questions



If conversational interviewing improves comprehension:

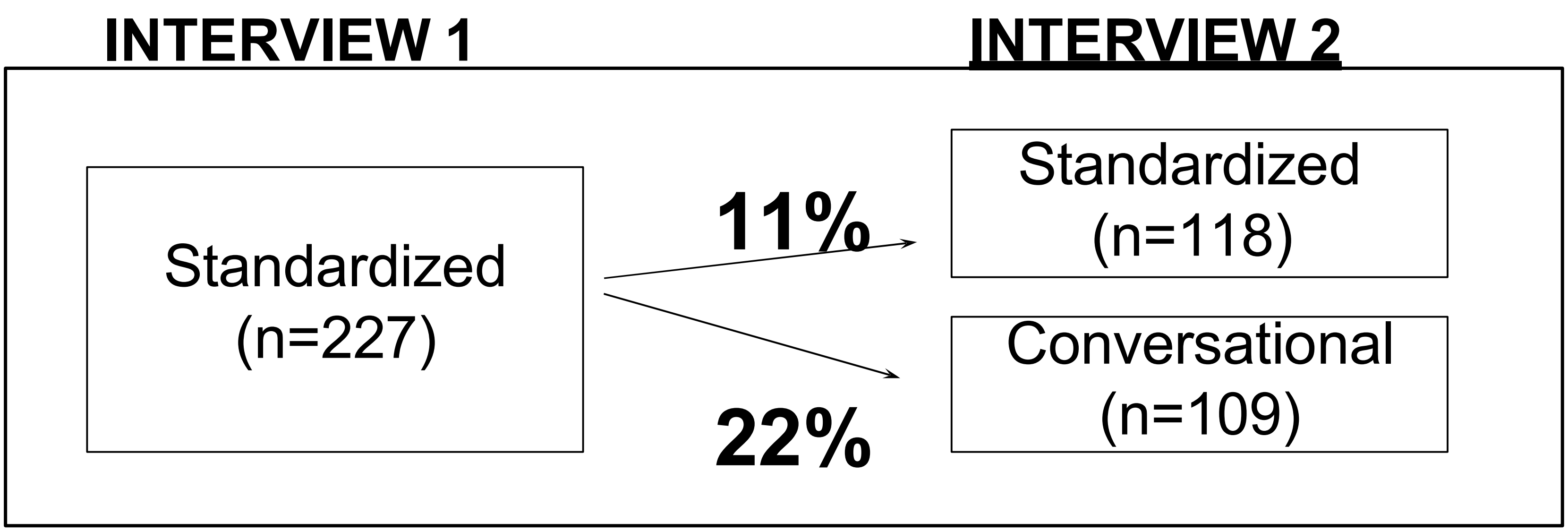
- Responses should change more from Interview 1 to Conversational Interview 2 than to Standardized Interview 2

INTERVIEW 1 **INTERVIEW 2**



Results: Response Change

- More responses changed when second interview was conversational than standardized



Median interview duration

Interview 1 (Standardized)	Interview 2	
5 mins	5 mins	Standardized
	9 mins	Conversational

Speed-Accuracy Tradeoff

- Strictly standardized interviews save time and therefore money
- Practitioners need to decide whether
  - it is worth the extra expense to be confident that comprehension is correct and uniform across *R*s
  - or whether some uncertainty about comprehension is tolerable
- But if time to recruit *R*s, which can take hours or days, is considered then increase in minutes due to conversational interviewing (even if threefold) negligible by comparison

Summary

- *R*s may misinterpret questions because they reach incorrect conclusions about literal meaning or make unwarranted inferences about intended meaning
- Literal meaning
  - Particular words may have different meaning for different *Rs*
  - how to map *R'*s circumstances to familiar concepts may be ambiguous
  - Grounding can help resolve such ambiguities in everyday conversation but not possible in standardized interviews
- Inferred meaning
  - *R*s transfer everyday inferential practices to survey response process leading, for example, to elimination of question order effects
  - *R*s may assume, based on cooperative principle, they are intended to consider info such as sponsorship

Summary (2)

- Conversational interviewing improves interpretation and response accuracy but increases interview duration
  - No evidence that *Iwer*s (mis)lead *R*s
  - Effective in both lab and field settings